Vladislav Makarov



2 March 2003

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EDUCATION

Master's degree — Moscow Institute of Physics and Technology (MIPT) 2025–2027
Phystech School of Applied Mathematics and Informatics
Degree programme: 09.04.01 Informatics and Computer Engineering
Department: Fundamental Methods of Artificial Intelligence
Average grade: 8.47 / 10
Thesis supervisor: Dmitry Yudin, Ph.D. in Engineering
Thesis topic: Development of a monolithic vision-language model for image-based 3D scene graph generation

Bachelor's degree — MIPT 2021–2025
Phystech School of Biological and Medical Physics
Degree programme: 19.03.01 Biotechnology
Department: Bioinformatics and Systems Biology
Average grade: 7.77 / 10
Thesis supervisor: Olga Kudryashova, Ph.D. in Biology
Thesis topic: Analysis of wide-field optical neuroimaging methods using machine learning and computer vision

Presidential Physics and Mathematics Lyceum No. 239 2017–2021

WORK EXPERIENCE

Intelligent Transport Laboratory, Center for Cognitive Modeling, MIPT August 2025 — Present
Position: Intern
Sientific supervisor: Dmitry Yudin

Federal Center of Brain Research and Neurotechnologies (FCBRN), FMBA of Russia September 2024 — July 2025
Position: External research staff member
Scientific supervisor: Olga Kudryashova

National Medical Research Center for Children's Health September 2024 — July 2025
Position: External research staff member

RESEARCH PROJECTS

Huawei July 2026 — Present
Improving video-based visual-spatial question-answering using 3D scene graphs.

Sber.Robotics June 2026 — August 2026
Development of a methodology for automatically creating training data for vision-language models that generate humanoid robot actions

Sber.Robotics August 2025 — December 2025
Development of intelligent methods for constructing and updating knowledge maps and their application to mobile manipulator localisation

RESEARCH PUBLICATIONS

1. **Makarov V.**, Gizetdinov M., Yudin D. SceneGraphVLM: Dynamic Scene Graph Generation from Video with Vision-Language Models. *arXiv:2605.13667*, 2026. arXiv
2. Pinkin E., Kartashov G., Kalmykov M., **Makarov V.**, Melekhin A., Yudin D. GraphSeqLoc: Image Sequence-Based Place Recognition via Scene Graphs and Temporal Re-Ranking. *SSRN 7422535*, 2026. SSRN

CONFERENCES

67th All-Russian Scientific Conference of MIPT April 2025
Track: Biological and Medical Physics
Session: Bioinformatics and Systems Biology
Presentation title: Bioinformatic analysis of wide-field optical neuroimaging data to investigate neurovascular changes in ischaemic stroke
Result: Winner

All-Russian Conference "Neurocampus 2025" April 2025
Interdisciplinary approaches to the study and treatment of nervous system diseases
Format: Poster presentation
Presentation title: Analysis of neurovascular changes in ischaemic stroke using wide-field optical neuroimaging data

PROFESSIONAL DEVELOPMENT

Advanced Machine Learning Methods February 2023 — December 2024
MIPT Digital Department — Diploma of Professional Retraining

TEACHING

Teaching Assistant — Phystech@DataScience, MIPT September 2024 — June 2025
MIPT · Statistics and Machine Learning for Applied Problems
Led tutorials in statistics and machine learning

HACKATHONS

SIBUR — "Element 119" AI in Chemistry Competition April 2025
Task: Prediction of the lipophilicity of organic molecules
Researcher · ML Engineer · Participating team · GitHub

BostonGene December 2024
Task: Analysis of immune repertoires in response to a pathogen by matching T-cell receptor CDR3 regions to known sequences in a database
Researcher · ML Engineer · **Winning team** · GitHub

MIPT Digital Department — Gazprom Neft November 2023 — December 2023
Task: Development of a web service for the automatic grouping of procurement requests for materials and equipment into lots
Team Captain · ML Engineer · Participating team

CASE COMPETITIONS

Phystech Business Solutions — Yakov and Partners November 2023 — December 2023
Task: Tripling the number of unique patients attending annual check-ups across the company's clinic network within five years
High-tech business · Researcher · Data Analyst · Participating team · Project materials

Phystech Business Solutions — Kept November 2022 — December 2022
Task: Development and justification of initiatives to optimise the business functions of an oil-refining holding company without changing production processes, with the aim of improving company performance
Researcher · Data Analyst · **Finalists, internship invitation** · Project materials

Phystech Business Solutions — Zerion April 2022 — May 2022
Task: Development of Web 3.0 solutions for secure P2P services that enhance trust, transparency and accessibility in economic interactions
Researcher · Data Analyst · **Finalists, internship invitation** · GitHub

INTERNSHIPS

Laboratory of Molecular Medical Diagnostics, MIPT July 2022 — August 2022

Intern · Developed professional skills in mass spectrometry

Infochemistry Scientific Center, ITMO University November 2019

Intern · Developed professional skills in infochemistry and materials science

KEY SKILLS

Languages and Frameworks	Python, PyTorch, Lightning, NumPy, Pandas, SciPy, scikit-learn, OpenCV, scikit-image, Seaborn, Matplotlib
Tools	Linux, Bash, Git, Docker, Jupyter, Isaac Sim, Rerun, CVAT, Codex, Cursor
Specialisation	Deep Learning, Computer Vision, Vision-Language Models (VLM), 3D Scene Graphs
Language Proficiency	Russian — Native · English — B1

REFERENCES

Dmitry Yudin
MIPT, Ph.D. in Engineering, Head of the Intelligent Transport Laboratory, Center for Cognitive Modeling and Scientific and Design Bureau of Computing Systems

Elizaveta Chefanova
MIPT, Team Lead, Product Analytics, Yandex

Olga Kudryashova
FCBRN, FMBA of Russia, Ph.D. in Biology, Head of the Bioinformatics and Omics Technologies Group

Radoslav Neychev
MIPT, Ph.D. in Computer Science, Co-developer of the master's programme "Modern Methods of Artificial Intelligence", Founder of girafe-ai

Daria Novikova
Saint Petersburg State University, Senior Lecturer, Department of Foreign Languages for Physics, Mathematics and Chemistry